

5.2.7 Lesson Review

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An attacker uses an AI tool to automatically rewrite malicious code so that it looks different each time, while still performing the same harmful actions. Signature-based scanners have trouble recognizing it as a known threat.

Which AI-related technique is being used in this scenario?

☒ Obfuscation ✔ Correct

☐ Automated exploitation

☐ Adversarial training

☐ Data correlation

Explanation

Obfuscation is the correct answer. Obfuscation changes how code appears without changing what it does. By constantly altering the code's form, the attacker makes it harder for signature-based tools to recognize it.

Adversarial training is incorrect. Adversarial training is used to harden models by training them on manipulated inputs. It does not describe repeatedly rewriting code to evade signatures.

Data correlation is incorrect. Data correlation links information across sources, such as matching usernames or credentials. It does not involve modifying code to hide its purpose.

Automated exploitation is incorrect. Automated exploitation focuses on finding and using vulnerabilities. While it may deliver payloads, the specific act of disguising code structure is better described as obfuscation.

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5.2.1 AI for Deception and Social Engineering

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A security operations analyst notices that a new strain of malware is bypassing the organization's antivirus solution, even though the vendor updates signatures several times a day.

Forensic analysis shows that each sample of the malware has the same behavior and goals but slightly different code structure, function names, and embedded strings. Network logs also indicate that the malware periodically changes its communication pattern in response to firewall rule updates.

Which scenario BEST explains how attackers are using AI to enable this malware campaign?

- ☐ An AI-driven framework continuously monitors threat intelligence feeds and security vendor signatures, then refactors the malware's code and command-and-control behavior based on those inputs, generating new variants that behave the same but look different to traditional detection tools.
- ☐ An AI-assisted toolkit ingests each blocked sample, learns which code features and network patterns triggered detection, and then regenerates slightly modified malware that preserves the same capabilities while altering signatures and communication profiles to slip past updated defenses.
- ☒ An AI model automatically rewrites and obfuscates the malware code to produce new variants that maintain the same functionality while evading signature-based detection, and adapts its behavior based on observed defensive responses. ✓ Correct
- ☐ An AI-powered malware builder uses reinforcement learning to test thousands of code and configuration permutations in a simulated environment, selecting variants that both retain core malicious behavior and avoid triggering antivirus and firewall rules, then periodically redeploys updated versions into the target environment.

Explanation

That an AI model automatically rewrites and obfuscates the malware code to produce new variants that maintain the same functionality while evading signature-based detection, and adapts its behavior based on observed defensive responses is the correct answer. This answer best fits the scenario because it directly accounts for both code-level variation and real-time behavioral adaptation to firewall changes. It describes AI that continuously rewrites and obfuscates the malware while adjusting its communication patterns based on actual defensive responses.

That an AI-assisted toolkit ingests each blocked sample, learns which code features and network patterns triggered detection, and then regenerates slightly modified malware is incorrect. This option focuses on learning only from blocked samples. The scenario shows malware largely evading detection from the outset, not primarily reacting after frequent failures. It underplays the proactive, continuous adaptation implied in the question.

That an AI-driven framework continuously monitors threat intelligence feeds and security vendor signatures is incorrect. Here, the AI reacts mainly to external threat intelligence and vendor signatures. The scenario instead points to fine-grained adaptation inside the victim environment, where communication changes follow specific firewall updates, not just public indicator changes.

That an AI-powered malware builder uses reinforcement learning to test thousands of code and configuration permutations in a simulated environment is incorrect. This describes powerful pre-deployment optimization in a simulated environment with periodic updates. The scenario suggests more immediate, in-environment adjustments to live firewall rule changes, which this periodic, lab-based approach does not fully capture.

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5.2.4 AI for Automated Attacks

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Which technique are attackers MOST likely to target with AI in order to distinguish fake systems from genuine production systems and avoid wasting effort?

☒ Honeypots ✓ Correct

- ☐ Data loss prevention (DLP) solutions
- ☐ Security orchestration, automation, and response (SOAR) platforms
- ☐ Vulnerability scanners

Explanation

Honeypots is the correct answer. Honeypots are decoy systems that imitate real assets to attract attackers. AI can help attackers analyze response timing, service behavior, and data consistency to spot signs that a target is a trap, allowing them to avoid honeypots and focus on genuine systems.

Security orchestration, automation, and response (SOAR) platforms is incorrect. SOAR platforms coordinate and automate security workflows rather than posing as attractive targets. Attackers may try to evade SOAR-driven responses, but they do not usually analyze SOAR systems themselves to decide whether they are fake or real environments.

Vulnerability scanners is incorrect. Vulnerability scanners are assessment tools, not systems that pretend to be production hosts. Attackers might use or evade scanners, but they do not typically apply AI to decide whether a scanner is a fake target in the same way they would with decoy hosts.

Data loss prevention (DLP) solutions is incorrect. DLP solutions monitor and control data movement rather than posing as endpoints to be attacked. While attackers may try to bypass DLP, using AI to determine if DLP components are fake systems is not a common or relevant focus.

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5.2.4 AI for Automated Attacks

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A security operations engineer is investigating why the organization's new AI-based email filter is suddenly allowing through a wave of sophisticated spam messages.

The messages look different from each other, with slight changes in wording, spacing, and character substitutions, yet they consistently bypass the model's classification while still looking legitimate to users.

The engineer suspects that attackers are using AI to repeatedly generate and refine these messages based on how the filter responds, gradually learning what will be accepted as "normal" email traffic.

Which technology is the attacker MOST likely using to systematically generate and test these evasive messages against the organization's AI defenses?

- ☐ Manual content review teams
- ☐ Rule-based spam filters
- ☒ Adversarial networks ✓ Correct
- ☐ Traditional signature databases for malware

Explanation

Adversarial networks is the correct answer. Adversarial networks can use a generator and a detector to iteratively create content that slips past defenses. Attackers can train one model to produce spam and another to mimic the email filter, refining the messages until they reliably evade the organization's AI.

Rule-based spam filters is incorrect. Rule-based spam filters rely on fixed patterns and keywords. They do not generate or adapt content against another AI system, so they cannot explain the automated, evolving spam in this scenario.

Traditional signature databases for malware is incorrect. Traditional signature databases store known malicious patterns but do not create or test new messages. They are used for detection, not for generating adaptive spam that learns from an AI filter's behavior.

Manual content review teams is incorrect. Manual review teams can adjust messages over time, but they cannot quickly generate and test large volumes of small variations. The high-speed, iterative evasion described points to automated adversarial techniques instead of human tuning.

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5.2.1 AI for Deception and Social Engineering

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How does AI-based automated data correlation most significantly strengthen reconnaissance efforts?

- ☐ By enriching open source intelligence with internal ticketing and asset records to build detailed profiles of an organization's critical systems.
- ☐ By clustering user activity logs and access records to reveal unusual relationships between accounts, roles, and business units.
- ☐ By aggregating network scan results with system inventory data to highlight exposed devices and services that match known attack patterns.
- ☒ By rapidly combining information from multiple public and leaked data sources to identify high-value targets and likely attack paths. ✓ Correct

Explanation

Rapidly combining information from multiple public and leaked data sources to identify high-value targets and likely attack paths is the correct answer. This option captures the main advantage of AI in reconnaissance: quickly merging many public and leaked data sources to pinpoint valuable targets and the most likely ways to reach them.

Aggregating network scan results with system inventory data is incorrect. This focuses on correlating scan results with asset data inside an environment, which is useful but narrower than broad external and leaked data correlation emphasized in the correct answer.

Clustering user activity logs and access records is incorrect. This describes behavioral analytics and insider or anomaly detection, which can support security, but it is less about large-scale external reconnaissance across diverse data sources.

Enriching open source intelligence with internal ticketing and asset records is incorrect. This improves understanding of internal critical systems, yet it lacks the wide-ranging combination of public and leaked data used to prioritize high-value targets and attack paths.

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5.2.3 AI for Reconnaissance and Data Correlation

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What is a key goal of using deepfake disinformation in a long-term social engineering campaign?

- ☐ To influence election outcomes by circulating synthetic clips that falsely show candidates making controversial statements.
- ☒ To intentionally manipulate public opinion by repeatedly spreading fabricated videos of recognizable individuals. ✓ Correct
- ☐ To sway consumer behavior by promoting realistic product reviews that slightly exaggerate real customer feedback.
- ☐ To shape attitudes within specific online communities by introducing edited and AI-enhanced recordings that appear to reveal hidden agendas.

Explanation

Intentionally manipulating public opinion by repeatedly spreading fabricated videos of recognizable individuals is the correct answer. Deepfake disinformation deliberately fabricates realistic videos of known individuals and distributes them over time to change how people view those individuals and related issues. This long-term manipulation of public opinion is a primary goal in extended social engineering campaigns.

Influencing election outcomes by circulating synthetic clips that falsely show candidates making controversial statements is incorrect. Targeting elections with false candidate clips is a form of manipulation, but it centers on a specific event and outcome. The correct answer describes a broader, ongoing effort to influence public opinion across many situations, which better reflects a long-term campaign.

Swaying consumer behavior by promoting realistic product reviews that slightly exaggerate real customer feedback is incorrect. Exaggerated product reviews can mislead consumers, yet this is closer to deceptive marketing than strategic deepfake disinformation. It does not clearly involve fully fabricated personas or large-scale opinion shaping like the scenario in the correct answer.

Shaping attitudes within specific online communities by introducing edited and AI-enhanced recordings that appear to reveal hidden agendas is incorrect. Misleading niche communities with edited, AI-enhanced recordings can shift some attitudes, but it suggests selective editing more than fully synthetic impersonation. The correct answer more directly captures the sustained use of realistic fake videos of recognizable individuals to broadly manipulate public perception.

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5.2.1 AI for Deception and Social Engineering

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Which description BEST explains how AI-based data correlation helps attackers use breached credential dumps more effectively?

- ☐ It archives old breach data and deletes duplicate entries to reduce storage and simplify backup procedures for the attacker.
- ☐ It replaces all stolen credentials with randomly generated strings to test password policy strength in a simulated environment.
- ☒ It compares usernames and passwords across multiple dumps to infer likely password reuse on other platforms. ✓ Correct
- ☐ It encrypts each breach file separately so that defenders cannot easily analyze stolen credentials.

Explanation

Comparing usernames and passwords across multiple dumps to infer likely password reuse on other platforms is the correct answer. AI-based data correlation can quickly compare large sets of breached credentials, detect repeated usernames and password patterns, and suggest which passwords are most likely reused across different services. This transforms raw breach data into prioritized, actionable intelligence for further attacks.

Encrypting each breach file separately so that defenders cannot easily analyze stolen credentials is incorrect. Encrypting breach files may protect them from interception, but it does not correlate or analyze the contents. It is a storage and secrecy measure, not a process that extracts patterns or relationships from multiple data sources.

Archiving old breach data and deleting duplicate entries is incorrect. Archiving and deduplication address data management, not analytic value. While they might save space, they do not reveal cross-dump patterns or identify which credentials provide useful paths to additional accounts.

Replacing all stolen credentials with randomly generated strings is incorrect. Replacing real credentials with random strings destroys the meaningful relationships within the data. This prevents the very pattern recognition and cross-referencing that data correlation relies on to support targeted attacks.

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 5.2.3 AI for Reconnaissance and Data Correlation

A regional bank's cybersecurity analyst notices a rapid spike in customer calls after a video goes viral on social media. The video appears to show the bank's CEO announcing that the bank's online systems were "permanently compromised" and advising customers to urgently move their funds to "temporary safe accounts" using account details shown at the bottom of the video.

The analyst is aware that no such breach has been reported internally, and the CEO is currently traveling.

Which action BEST applies security practices to address this apparent use of AI-generated misinformation?

- ☒ Escalate the incident, verify the video with corporate communications and the CEO's office, coordinate an official statement, and work with platform contacts to report and remove the fake content. ✔ Correct
- ☐ Verify with the CEO's assistant that no such announcement was made, log the incident, and post a short clarification on the bank's official social media accounts.
- ☐ Quickly draft an email to customers stating that the bank is investigating the video, advising them not to move funds, but delay contacting social media platforms or leadership until technical checks are complete.
- ☐ Confirm with the incident response team that systems show no evidence of compromise, then publish a customer advisory on the website and mobile app while allowing the video to remain online.

Explanation

Escalating the incident, verifying the video with corporate communications and the CEO's office, coordinating an official statement, and working with platform contacts to report and remove the fake content is the correct answer. This option verifies the message, coordinates an official response, and works to remove the fake video. It directly counters deepfake misinformation and limits both its spread and impact on customers.

Verifying with the CEO's assistant that no such announcement was made, logging the incident, and posting a short clarification is incorrect. This response verifies the message but keeps the effort narrow and informal. Without broader coordination or takedown efforts, the deepfake can continue to circulate and confuse customers.

Confirming with the incident response team that systems show no evidence of compromise is incorrect. Confirming there is no breach and posting an advisory helps, but leaving the video online gives the misinformation continued visibility, so customers may still trust and follow the fake instructions.

Quickly drafting an email to customers stating that the bank is investigating the video is incorrect. Alerting customers is useful, but delaying escalation to leadership and platforms slows the overall response. While the email goes out, the deepfake continues to spread without coordinated containment.

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5.2.1 AI for Deception and Social Engineering

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How does AI most directly help malware evade traditional signature-based detection tools?

- ☒ By continuously reshaping the malware's code through automated obfuscation and minor variations while keeping its behavior similar. ✓ Correct
- ☐ By analyzing antivirus logs to schedule malware execution during low-activity periods when scanning is less frequent or more predictable.
- ☐ By learning which file formats and packaging methods are less likely to trigger alerts and wrapping the malware accordingly for delivery to targets.
- ☐ By automatically selecting from different prebuilt malware modules so that each infection uses a slightly different feature set and execution path.

Explanation

Continuously reshaping the malware's code through automated obfuscation and minor variations while keeping its behavior similar is the correct answer. AI can automatically generate many code-level variants of the same malware, changing structure and appearance while preserving function. This ongoing obfuscation makes static signatures that look for specific patterns far less reliable.

Automatically selecting from different prebuilt malware modules is incorrect. Varying prebuilt modules changes features and behavior, but the modules themselves can still match known signatures. AI-driven evasion is more focused on rewriting or transforming code so that underlying patterns no longer match existing signature rules.

Analyzing antivirus logs to schedule malware execution during low-activity periods is incorrect. Adjusting execution timing may reduce attention from analysts but does not change how signatures work. Signature-based tools will still flag known code patterns whenever scanning occurs, so timing alone does not directly defeat this type of detection.

Learning which file formats and packaging methods are less likely to trigger alerts is incorrect. Choosing less suspicious file formats or packing methods can help initial delivery but not the core signature problem. Once the malware is unpacked or runs in memory, its code is still examined, and unchanged patterns can still be matched by existing signatures.

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 5.2.4 AI for Automated Attacks

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A threat intelligence analyst is reviewing unusual login attempts against a company's cloud portal. The attempts originate from many different IP addresses but share a common pattern: each attempt uses a slightly different username and password combination that closely resembles credentials leaked in several unrelated data breaches over the last five years.

Further investigation shows that an attacker is using an AI tool that rapidly aggregates old breach dumps, correlates reused passwords, and prioritizes likely account targets inside the company.

Which action by the attacker BEST demonstrates the use of AI for reconnaissance with automated data correlation in this scenario?

- ☒ Using AI to crawl public breach databases, cross-reference credential sets, and generate a prioritized list of likely valid usernames and passwords for the company's accounts. ✓ Correct
- ☐ Using AI-powered bots to flood the cloud portal with a massive volume of simultaneous connection requests, overwhelming its resources and making it unavailable to legitimate users.
- ☐ Using AI to automatically modify malware code in real time to evade antivirus signatures and behavior-based detection engines deployed on company endpoints.
- ☐ Using AI to generate a highly convincing spear-phishing email that mimics the writing style of the company's CEO and includes a link to a fake single sign-on page.

Explanation

Using AI to crawl public breach databases, cross-reference credential sets, and generate a prioritized list of likely valid usernames and passwords for the company's accounts is the correct answer. This is AI-enabled reconnaissance because it gathers information from many breach dumps, correlates reused credentials, and produces a prioritized target list. The focus is on discovering likely valid accounts before any direct exploitation.

Using AI to generate a highly convincing spear-phishing email that mimics the writing style of the company's CEO is incorrect. This is AI-assisted social engineering. The AI is used to craft a realistic phishing email to trick users, which is part of delivery and exploitation, not broad data gathering and correlation across multiple sources.

Using AI to automatically modify malware code in real time is incorrect. This describes AI-assisted malware evasion. Modifying code to bypass antivirus is about payload stealth during execution, not about collecting and correlating target information in the reconnaissance phase.

Using AI-powered bots to flood the cloud portal with a massive volume of simultaneous connection requests is incorrect. This is an AI-enhanced DDoS attack. AI helps overwhelm the service to disrupt availability, but it does not involve profiling users or correlating data about potential targets.

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5.2.3 AI for Reconnaissance and Data Correlation

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